ANX-PR/CL/001-01
LEARNING GUIDE

SUBJECT

303000083 - Ecuaciones Diferenciales Ordinarias Y Aplicaciones

DEGREE PROGRAMME

30AH - Máster Universitario En Matemáticas Avanzadas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

Index

Learning guide

1. Description

1.1. Subject details

Name of the subject	303000083 - Ecuaciones Diferenciales Ordinarias y Aplicaciones
No of credits	6 ECTS
Type	Optional/elective
Academic year ot the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	30AH - Máster Universitario en Matemáticas Avanzadas
Centre	30 - Esc. Politéc. Enseñanza Superior (epes)
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Salvador Jimenez Burillo (Subject coordinator)	A-124 (ETSIT)	s.jimenez@upm.es	W - 15:00 - 16:00

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Basic knowledge on ordinary differential equations and/or dynamical systems

- Highly recommended: knowledge of Maple, Matlab, C, Python.

- English language

4. Skills and learning outcomes *

4.1. Skills to be learned

CO1 - Conocer los principales métodos y teorías matemáticas, próximos a la frontera del conocimiento, que permitan tener una amplia perspectiva en al menos tres de las siguientes áreas: Análisis Matemático, Métodos Numéricos, Ecuaciones Diferenciales, Estadística y Ciencia de Datos, Álgebra, Geometría y Topología. TIPO: Conocimientos o contenidos.

CO2 - Conocer demostraciones matemáticas avanzadas que permitan relacionar de una manera rigurosa las hipótesis con las conclusiones en al menos tres de las áreas relacionadas en el CO1.TIPO: Conocimientos o contenidos.

CO3 - Conocer diversas aplicaciones de las teorías matemáticas y su modelización en otros entornos o disciplinas que permitan adaptar los conocimientos matemáticos a nuevos problemas.TIPO: Conocimientos o contenidos.

CP1 - Utilizar programas informáticos especializados como laboratorio para diseñar, construir o validar hipótesis científicas en.alguno de los ámbitos de las matemáticas. TIPO: Competencias.

CP2 - Ser capaz de trabajar en equipos multidisciplinares que involucren conocimientos matemáticos avanzados en áreas teóricas o aplicadas.TIPO: Competencias.

CP3 - Seleccionar y comprender la bibliografía científica sobre un tema concreto, extrayendo la información relevante. TIPO:Competencias.

CP4 - Comunicar de forma oral y escrita conocimientos y conclusiones científicas, valorando diferentes interpretaciones en públicos especializados y no especializados de una manera clara y rigurosa. TIPO: Competencias

CP5 - Complementar la formación matemática de manera autónoma en avances específicos de las Matemáticas, seleccionando la bibliografía adecuada y planteando cuestiones de interés. TIPO: Competencias

HA1 - Aplicar los conocimientos matemáticos adquiridos a la resolución de problemas nuevos de una manera crítica, generando hipótesis de trabajo claras, definiciones precisas, nuevos enfoques con fundamento matemático y soluciones originales. TIPO: Habilidades o destrezas

HA2 - Ser capaz de abstraer y detectar las dificultades relevantes ante problemas complejos para poder plantearlos correctamente y resolverlos con técnicas matemáticas avanzadas. TIPO: Habilidades o destrezas

HA3 - Dominar el rigor matemático en las técnicas de demostración aprendidas para su posterior uso en el enunciado y demostraciones de nuevos teoremas. TIPO: Habilidades o destrezas

HA4 - Poseer destreza para proponer, validar e interpretar modelos matemáticos de diferente dificultad en situaciones reales complejas, utilizando las herramientas matemáticas más adecuadas a los fines que se persigan, y valorando las hipótesis y conclusiones asociadas a cada nivel de dificultad del modelo. TIPO: Habilidades o destrezas

HA5 - Adaptar teorías y técnicas matemáticas avanzadas al estudio de problemas en otros ámbitos con criterio científico para valorar su repercusión y formular las hipótesis adecuadas. TIPO: Habilidades o destrezas

HA6 - Interpretar adecuadamente la relevancia de nuevos resultados en Matemáticas, su aplicabilidad en situaciones particulares donde pudieran suponer nuevos avances y su relación con los resultados existentes. TIPO: Habilidades o destrezas

4.2. Learning outcomes

RA21 - Entender el análisis de estabilidad para modelos dinámicos

RA22 - to be able to analyse and describe the behaviour of the solutions of ordinary differential equations that arise from modelling and applications, using both analytical and computational tools

RA5 - Desarrollar trabajos autónomos o en grupo, integrando bibliografía científica y, en su caso, métodos computacionales para profundizar en los contenidos de la asignatura o abordar problemas complejos.

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Description: this course aims to provide the student with the necessary tools to analyse and describe the behaviour of the solutions of ordinary differential equations that arise from modelling and applications. The tools are analytic, on one part, but also numeric.

Previous knowledge: the student is supposed to be familiar with the theory of ODEs as well as the qualitative behaviour of dynamical systems. Those students that come from the degree in mathematics (Grado en Matemáticas) will have acquired this in Ecuaciones Diferenciales Ordinarias I and II. Also a minimum familiarity with numerical tools will be expected. Again, those students from the degree in mathematics will have acquired this knowledge in Cálculo Numérico I and, especially, II.

Links to other master courses: this course will provide in a sense a complementary practical or applied vision of the dynamical systems theory and topics that correspond to ?Sistemas dinámicos avanzados S1-I?. The fact that both will be taught on the first semestrer may enhance a combined complementarity.

5.2. Syllabus

1. Introduction and fundamentals
    1.1. Review of basic properties of ODE and IVP. Existence, unicity and regularity.
    1.2. Review of linear autonomous systems.
    1.3. Periodic linear systems. Floquet theory.
    1.4. Singular perturbative problems and multiple-time expansion.
    1.5. Review of numerical methods. Stiff equations.
2. Nonlinear systems
    2.1. Trayectories. Phase diagrams. Poincaré and stroboscopic maps.
    2.2. Critical points and Lyapunov stability.
    2.3. Periodic orbits and limit cycles. Stability.
    2.4. Invariant manifolds. Attraction domains.
3. Bifurcations and chaos
    3.1. Structural stability and bifurcations.
    3.2. Characterisation of chaos. Lyapunov exponents, Lagrangian descriptors, Melnikov method.
    3.3. Chaotic models. Intermittency and the onset of chaos.
4. Applications
    4.1. Introduction to control of nonlinear systems.
    4.2. Coupled oscillators.
    4.3. Growth models in biology
    4.4. Particle trapping.

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	**topics 1.1 to 1.3** Duration: 03:00 Lecture **topics 1.1 to 1.3** Duration: 01:00 Problem-solving class			
2	**topics 1.3 and 1.4** Duration: 03:00 Lecture **topics 1.3 and 1.4** Duration: 01:00 Problem-solving class			
3	**topic 1.5** Duration: 01:00 Lecture **topic 1.5** Duration: 01:00 Problem-solving class **topic 2.1** Duration: 01:00 Lecture **topic 2.1** Duration: 01:00 Problem-solving class			
4	**topics 2.1 and 2.2** Duration: 03:00 Lecture **topic 2.2** Duration: 01:00 Laboratory assignments			
5	**topic 2.3** Duration: 03:00 Lecture **topic 2.3** Duration: 01:00 Problem-solving class			

6	**topic 2.3** Duration: 01:00 Lecture **topic 2.3** Duration: 01:00 Problem-solving class **topic 2.4** Duration: 02:00 Lecture			
7	**topic 2.4** Duration: 03:00 Lecture **topic 2.4** Duration: 01:00 Problem-solving class			**Handout of a lab and problem related to the first two chapters** Group work Progressive assessment Not Presential Duration: 00:00
8	**topic 2.4** Duration: 01:00 Lecture **topic 2.4** Duration: 01:00 Problem-solving class **topic 3.1** Duration: 02:00 Lecture			
9	**topic 3.1** Duration: 01:00 Lecture **topic 3.1** Duration: 01:00 Problem-solving class **topic 3.2** Duration: 02:00 Lecture			
10	**topic 3.2** Duration: 03:00 Lecture **topic 3.2** Duration: 01:00 Problem-solving class			
11	**topic 3.2** Duration: 02:00 Lecture **topic 3.2** Duration: 01:00 Problem-solving class **topic 3.3** Duration: 01:00 Lecture			

12	**topic 3.3** Duration: 03:00 Lecture **topic 3.3** Duration: 01:00 Problem-solving class			
13	**topic 3.3** Duration: 02:00 Lecture **topic 4.1** Duration: 02:00 Lecture			**Handout of a lab and problem related to the third chapter. It will be a continuation of the models treated in the previous handout.** Group work Progressive assessment Not Presential Duration: 00:00
14	**topic 4.2** Duration: 02:00 Lecture **topics 4.1 and 4.2** Duration: 01:00 Problem-solving class **topic 4.3** Duration: 01:00 Lecture			
15	**topics 4.3 to 4.4** Duration: 03:00 Lecture **topic 4.4** Duration: 01:00 Problem-solving class			
16				**Handout of a group project based on an application** Group work Progressive assessment Not Presential Duration: 00:00
17				**Written exam** Written test Progressive assessment Presential Duration: 03:00 **Written exam - ordinary period. For those that do not perform the progressive evaluation** Written test Global examination Presential Duration: 03:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
7	Handout of a lab and problem related to the first two chapters	Group work	No Presential	00:00	20%	3 / 10	CO1 CO2 CO3 CP1 CP2 CP3 CP4 CP5 HA1 HA2 HA3 HA4 HA5 HA6
13	Handout of a lab and problem related to the third chapter. It will be a continuation of the models treated in the previous handout.	Group work	No Presential	00:00	20%	3 / 10	CO1 CO2 CO3 CP1 CP2 CP3 CP4 CP5 HA1 HA2 HA3 HA4 HA5 HA6
16	Handout of a group project based on an application	Group work	No Presential	00:00	30%	3 / 10	CO1 CO2 CO3 CP1 CP2 CP3 CP4 CP5 HA1 HA2 HA3 HA4

							HA5 HA6
17	Written exam	Written test	Face-to-face	03:00	50%	3 / 10	

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Written exam - ordinary period. For those that do not perform the progressive evaluation	Written test	Face-to-face	03:00	100%	5 / 10	CO1 CO2 CO3 CP1 CP2 CP3 CP4 CP5 HA1 HA2 HA3 HA4 HA5 HA6

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Written exam - extraordinary period	Written test	Face-to-face	03:00	100%	5 / 10	CO1 CO2 CO3 CP1 CP2 CP3 CP4 CP5 HA1 HA2 HA3 HA4 HA5

							HA6

7.2. Assessment criteria

The evaluation will be graded according to the following criteria:

- the originality of the work,
- the good execution, exposition, reasoned achievement and completeness of the requested demonstrations or analyses,
- the proper use of the computational tools,
- the proper presentation of the results,
- the adequacy of the work to the specifications given in the statements.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
M. W. Hirsch, S. Smale and R. L. Devaney. Differential Equations, Dynamical Systems, and an Introduction to Chaos. Third edition, Elsevier, 2013, https://doi.org/10.1016/B978-0-12-382010-5.00057-9.	Bibliography	A general book to cover most of the course
C. Chicone. Ordinary Differential Equations with Applications. Texts in Applied Mathematics, vol 34. Springer, 2024, https://doi.org/10.1007/978-3-031-51652-8.	Bibliography	A general book and a very good reference for all students to follow.
E. Hairer, G. Wanner, S. P. Nørsett. Solving Ordinary Differential Equations I. Springer 1993, https://doi.org/10.1007/978-3-540-78862-1.	Bibliography	Covers all numerical methods with the exception of stiff equation.

E. Hairer, G. Wanner. Solving Ordinary Differential Equations II. Springer, 1996, https://doi.org/10.1007/978-3-642-05221-7.	Bibliography	numerical methods, specific to stiff problems.
E. Hairer, G. Wanner, Ch. Lubich. Geometric Numerical Integration. Springer, 2006 , https://doi.org/10.1007/3-540-30666-8.	Bibliography	A perfect reference book for symplectic or geometric numerical methods for problems with symmetries and conservation laws. Especially relevant for Hamiltonian systems.
A. Stuart, A.R. Humphries, Dynamical systems and numerical analysis, Cambridge, 1996.	Bibliography	A basic and complete book to simulate dynamical systems and their evolution equations.
L. Vázquez, S. Jiménez, C. Aguirre, P.J. Pascual. Métodos numéricos para la Física y la Ingeniería. McGrawHill, 2009.	Bibliography	A basic book that presents conservative numerical methods and its generalizations. Although it is in Spanish it can be used by non-Spanish readers with the help of the teacher.
L. Perko. Differential Equations and Dynamical Systems. Third edition. Springer, 2001, https://doi.org/10.1007/978-1-4613-0003-8.	Bibliography	Very complete book. Also, a very good reference for the structure of the phase space and the analysis of the stable, unstable and center manifolds. Topics 2 and 3 of the course.
S. Wiggins. Introduction to Applied Nonlinear Dynamical Systems and Chaos. Second edition. Springer, 2003, https://doi.org/10.1007/b97481.	Bibliography	Similar to the previous one (Perko) but perhaps a more demanding level.
D.K. Arrowsmith, C.M. PLace. An introduction to dynamical systems. Cambridge University Press, 1994.	Bibliography	A very good reference for bifurcations.
J. Hale, H. Koçak. Dynamics and Bifurcations. Springer, 1991, https://doi.org/10.1007/978-1-4612-4426-4.	Bibliography	This book is a classic on bifurcations. Although the programs that come with it are nowadays obsolete it remains a very interesting text.

E. Ott. Chaos in Dynamical Systems. Cambridge University Press, 2002, https://doi.org/10.1017/CBO9780511803260.	Bibliography	A general book that may also be used for the whole course. Especially interesting for the applications it presents.
S. Strogatz. Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry and Engineering. Second edition, Westview Press, 2015, https://doi.org/10.1201/9780429492563 .	Bibliography	Same as the previous one (Ott), especially interesting for all the applications it presents.
S. Lynch, Dynamical Systems with Applications using Maple. Second Edition, Birkhäuser 2010, https://doi.org/10.1007/978-0-8176-4605-9.	Bibliography	This book provides ready tools to help the analysis of the properties of the equations, using the symbolic programme Maple.
Other material provided by the teacher	Others	Since this is not a numerical methods course, supporting material such as programs to analyse or simulate will either be provided by the teacher or be of free distribution. Languages: C, Matlab, Maple and Python.
Salvador Jiménez	Others	Presentation notes and slides by the teacher.
Research papers	Bibliography	Some research papers, by different authors and on specific topics, provided by the teacher.

9. Other information

9.1. Other information about the subject

The planning described might be slightly affected by daily adjustments. There is not a clear distinction between theoretical and problem or practical classes since the idea is to have the theoretical presentations intermingled with practical implementations and exercises, side by side, perhaps inside the same session.